Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

1. Technology Proposer Information

Proposer	Name	Semih Akin (Principal Investigator) Nikhil Koratkar (Co-Principal Investigator)	Email	akins@rpi.edu
Institute Name	Rensselaer Polytechnic Institute			
Address	110 8th Street, Troy, NY 12180, USA			

2. Technology Overview

Title of Proposed Technology	Solvent-and Binder-Free Additive Manufacturing of Lithium-Ion Battery Electrodes			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical · Electronics		Electrical · Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Batteries	Secondary Batteries	Batteries	Battery Materials
	Technology Field 3 (Large Category)		**Technology Field 4** (Large Category)	
	Electrical · Electronics		Mechanical · Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Batteries	Manufacturing and Measurement	Plastic Deformation-Powder	Powder Processing Technologies
Concept of Proposed Technology	**Project overview:** In this project, as depicted in **Figure 1**, we propose a **manufacturing innovation** that leverages the **emerging Cold Spray Additive Manufacturing (CSAM) technique to fabricate fully dry-coated lithium-ion battery (LIB) electrodes.** • The proposed technology utilizes a high-velocity, dry-deposition process—specifically **supersonic cold spraying**—to manufacture LIB electrodes **within a single step**. • This method could eliminate the need for **toxic organic solvents (e.g., NMP** [1]**)** and **traditional polymer binders** [2], <u>significantly reducing the environmental footprint and cost of battery production.</u> • Given the solid-state nature of the CSAM process, it is expected to prevent the thermal degradation of active materials. This may result in superior **inter-particle connectivity, strong interfacial adhesion strength,** and **enhanced electronic conductivity** throughout the electrode matrix. **Process overview:** ➔ The core of the technology involves accelerating functional battery active materials to supersonic speeds and impacting them onto a current collector (target substrate). The kinetic energy of the particles is converted into interfacial bonding followed by mechanical interlocking upon impact, creating a dense, cohesive **"functional coating (i.e., electrode)"** without chemical additives (see **Figure 1**).			

(max 10 pages)

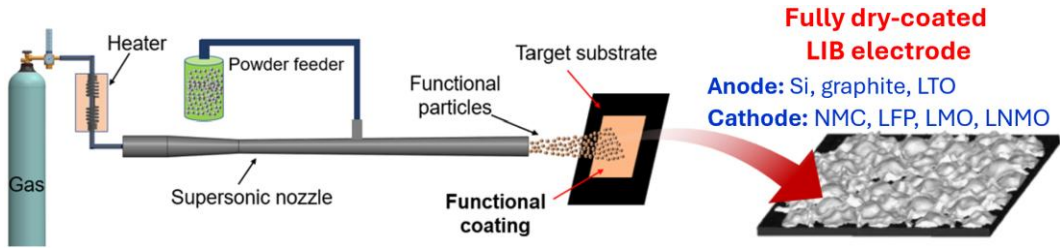

Figure 1: *Proposed manufacturing innovation for the fully dry-coated LIB electrodes.*

- ➔ In CSAM, fine particles (5-100 μm) are accelerated to supersonic velocities (300-1200 m s^{-1}) using a high-pressure gas (e.g., air, nitrogen, helium) [3-4]. Upon impact with a target substrate (current collector), severe plastic deformation occurs, resulting in a fully dry coating formed at low process temperatures with strong adhesion, while preserving the intrinsic properties of the feedstock powders [5-6].
- ➔ Compared with conventional deposition methods, CSAM offers high deposition rates, low porosity, and improved adhesion, all achieved under solid-state conditions with minimal thermal input [5]. Hence, these attributes make CSAM particularly well suited for producing fully dry-coated high-performance, durable LIB electrodes.

LIB electrode material versatility:

- ➔ This project will explore **fully dry coating of a range of industry-relevant LIB electrode materials, including but not limited to <u>silicon, graphite, silicon-graphite composites, tin and lithium titanate (LTO) as anode materials, as well as lithium nickel manganese cobalt oxide (NMC), lithium iron phosphate (LFP), lithium manganese oxide (LMO), lithium nickel manganese oxide (LNMO) and lithium nickel cobalt aluminum oxide (NCA) as cathode materials</u>.**
- ➔ In parallel, this work will establish clear process–structure–property–performance relationships linking CSAM processing parameters to electrode microstructure, electrochemical behavior, and long-term cycling stability.
- ➔ Furthermore, building on the most promising dry-coated electrode (anode and cathode) architectures, this proposal will aim to develop **fully dry-coated full-cell LIB architectures**.

Table 1. *Electrode material systems to be investigated in this proposal.*

Electrode Material	
Graphite (C)	→ *Anode*
Silicon (Si)	→ *Anode*
Tin (Sn)	→ *Anode*
Lithium iron phosphate ($LiFePO_4$)	→ *Cathode*
Lithium manganese oxide ($LiMn_2O_4$)	→ *Cathode*
Lithium nickel manganese cobalt oxide ($LiNi_xMn_yCo_{1-x-y}O_2$)	→ *Cathode*
Lithium nickel cobalt aluminum oxide ($LiNi_xCo_{1-x-y}Al_yO_2$)	→ *Cathode*

Preliminary results:

We deposited LTO anode and LMO cathode layers onto aluminum current collectors using CSAM, demonstrating its feasibility for fully dry-coated LIB electrode fabrication. All powders were sourced from a commercial supplier (*MSE Supplies Inc., USA*) and used as received. Half-cell battery tests were conducted to evaluate electrochemical performance of the fabricated electrodes. The electrodes were assembled with Li metal as the counter-electrode in coin cells. A Celgard 2340 polypropylene membrane was used as the separator and a ~1M LiPF6 in 1:1 mixture of ethylene carbonate and diethyl carbonate was used as electrolyte. An Arbin BT2000 Battery Tester was used to carry out galvanostatic charge/discharge cycling. **Figures 2a–b** benchmark CSAM-fabricated electrodes against recent solvent-free techniques, including hot pressing [7], dry pressing [8], electrostatic spraying [9], cold pressing [10], lamination [11,12], dry spraying [13], and cold plasma [14]. Unlike these studies that eliminate solvents but still rely on binders, the **CSAM approach is an entirely dry coating process**.

As shown **in Figure 2a**, the CSAM LTO anode exhibited stable electrochemical performance with no significant capacity fading over 400 cycles, outperforming the dry spraying + hot pressing method. <u>This stable cycling behavior positions the CSAM-based LTO anode as a strong contender within the existing body of solvent-free dry-coating strategies.</u>

For the cathode (**Figure 2b**), the CSAM LMO electrode showed electrochemical behavior comparable to other solvent-free methods, with cycling performance similar to conventional calendaring and state-of-the-art electrostatic spraying. Overall, the CSAM-deposited electrodes demonstrated promising cycling stability—minimal capacity fade, low voltage hysteresis, and highly reversible electrochemical behavior.

(max 10 pages)

Taken together, these preliminary results indicate that CSAM-deposited battery electrodes deliver promising cycling stability; minimal capacity fade and voltage hysteresis and highly reversible electrochemical performance, even after 300 cycles. These promising results stem from the intrinsic features of CSAM, including; (i) high-energy solid-state particle consolidation; (ii) strong interparticle and particle–current collector bonding; and (iii) the formation of fully-dry mechanically robust electrode architectures.

Potential applications: Rechargeable LIBs; green and sustainable energy storage; load-bearing LIBs; internet of things (IoT); portable electronics, etc.

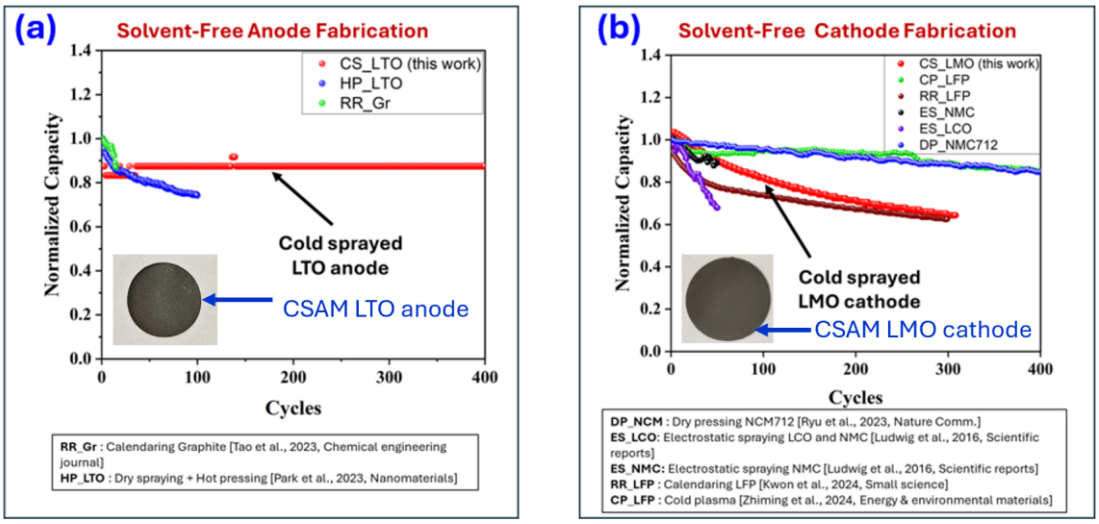

***Figure 2. Preliminary results and performance benchmarking** of solvent-free LIB **(a)** anode and **(b)** cathode fabrication methods [7-14]. Note that most of these studies eliminate solvents but still rely on binders; in contrast, the CSAM approach is a fully dry, binder-free coating process.*

Rationale for Joint R&D Project

A joint project is needed for four specific reasons:

- **CSAM of industry-relevant electrode materials**: Translation to CSAM-based LIBs require validation with commercially dominant chemistries, including silicon and graphite anodes and LFP and NMC cathodes.
- **Scale-up and cell-level validation:** Our current testing capabilities are primarily limited to coin-cell configurations. However, demonstrating proof-of-concept and assessing commercial relevance require validation in larger form factors, such as pouch or prismatic cells. Industry partners are uniquely positioned to lead and guide this scale-up effort, leveraging their expertise in advanced cell fabrication, integration, and testing.
- **Full-cell LIB development:** While our laboratories have strong capabilities in fabricating and characterizing individual CSAM-based anodes and cathodes, integrating these components into balanced, fully dry-coated full-cell LIB architectures requires industrial expertise in electrode pairing, electrolyte selection, and safety validation. A joint effort is therefore essential to demonstrate practical full-cell performance and manufacturability.
- **Cost and scalability analysis:** A rigorous assessment of manufacturing cost, throughput, and scalability is critical to evaluate the techno-economic viability of CSAM–based dry electrode manufacturing. Industry involvement is necessary to provide realistic process models, cost structures, and benchmarking against incumbent manufacturing routes.

Key Words

1. Lithium-ion batteries (LIBs)
2. Battery manufacturing innovation
3. Cold spray additive manufacturing (CSAM)
4. Dry-coating of LIB electrodes
5. Solvent-and binder-free electrodes

(max 10 pages)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)

R&D Objectives

The PIs of this proposal have demonstrated the **proof-of-the-concept validation** of the CSAM for LIB fabrication in a solvent- and binder-free manner, which corresponds to the TRL 3 for the developed approach. The **primary goal** of this joint project is to transition CSAM from **a laboratory-scale process to an industrially viable manufacturing platform** for solvent- and binder-free LIB electrodes. To this end, the proposed R&D effort will pursue the following objectives:

- **Solvent- and binder-free processing of LIB electrodes:** Establish and optimize CSAM parameters to produce high-quality dry-coated anode and cathode layers without the use of solvents or polymeric binders.
- **Establish process–structure–performance relationships** by correlating CSAM parameters and resulting microstructures with electrochemical and mechanical performance metrics.
- **Scaling to commercial form factors:** To transition from coin-cell prototypes to industry-relevant architectures (e.g., pouch and/or prismatic cell configurations)
- **Full-cell integration and optimization:** Design and validate fully dry-coated LIB full-cell architectures with (i) electrode pairing (i.e., balanced N/P ratios); (ii) electrolyte optimization; and (iii) performance testing (i.e., benchmarking against slurry-coated cells)
- **A techno-economic and scalability assessment** to quantify the commercial viability of CSAM-based manufacturing compared to wet-slurry casting.

Specific technologies to develop:

- An additive manufacturing process for fully dry-coated LIB electrodes
- Engineered electrode feedstock (powder) materials
- Validated process window for solid-state deposition of LIB electrodes
- Full-cell prototype validation and benchmarking

R&D Target Specifications

The proposed R&D efforts aim to establish quantifiable performance and manufacturing targets to guide the development of CSAM LIB electrode technology. Key target specifications include:

- High areal mass loading (>10 mg/cm²) with uniform deposition and minimal defects.
- Production of electrodes with high packing density to meet industry standards.
- Stepwise demonstration in **coin-cell** and **pouch-cell** formats, meeting predefined benchmarks for specific capacity, coulombic efficiency, operating voltage and energy density.
- Cycle stability over extended charge-discharge cycles (>1000 cycles with a capacity fade rate of < 0.01% per cycle).
- Fast C-rate charge acceptance (>1C) enabling fast-charging performance.
- Strong adhesion strength (>10 MPa) and durability under mechanical and thermal stresses typical of industrial handling and cell assembly.
- The cost and environmental footprint for LIB manufacturing by CSAM should be lower than that of conventional slurry-based LIB manufacturing.

R&D Plan

To achieve the R&D objectives, we propose a structured task plan as shown in **Figure 3.**

Task 1: Material feasibility & parameter optimization *(University - led)*

We will optimize particle size distribution, morphology, and flowability for industry-relevant active materials, including Si, graphite, NMC, NCA, LFP, LTO, and LMO. This includes powder sieving, blending, and flowability testing (Hall flow, tap density) to ensure consistent feeding during deposition. Where needed, powder conditioning methods (e.g., mechanical milling, thermal pre-treatment, surface functionalization) will be explored to enhance deformability and reduce brittleness. By engineering the feedstock at the particle level, we expect to overcome the traditional barriers to dry-coating of electrodes to meet industry-standard mass loadings without compromising structural integrity.

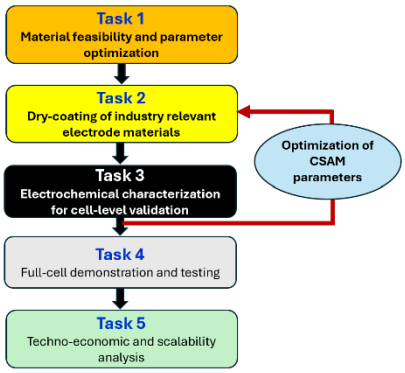

Figure 3. *Proposed plan of work.*

(max 10 pages)

Task 2: Dry coating of industry relevant electrode materials *(University - led)*

The objective of this task is to establish the process-structure-property relationships for CSAM across industry-relevant LIB electrode chemistries. We will aim to transition from single-layer coatings to high-performance, thick-film electrodes that match commercial specifications. To achieve this, we will develop multi-pass deposition strategies to build thick, cohesive electrodes while managing residual stress and preventing delamination. Specifically, we will leverage the compaction effect of supersonic cold spray impact to achieve dense electrodes that **bypass the traditional energy-intensive calendaring process to reduce manufacturing costs.**

The cold spray deposition experiments will be conducted in PI Akin's laboratory using the robotic CSAM cell (**Figure 4a)**. By optimizing the critical process parameters as shown in **Figure 4b**, we aim to achieve industry-relevant electrode properties, specifically targeting optimized mass loading, porosity, ductility, adhesion strength, and fracture toughness. For this, we will develop a design of experiments (DoE) to systematically investigate and optimize the CSAM process parameters for high-quality electrode fabrication.

Following the fabrication of the electrodes, comprehensive characterization will be conducted to verify that the binder-free electrodes meet industrial structural standards. Morphological and phase analysis will be conducted via Scanning Electron Microscopy (SEM) and X-ray Diffraction (XRD), while mechanical integrity will be evaluated through nano-indentation and pull-off adhesion testing.

Mitigation plan: If the target mass loading is not achieved, we will investigate **mechanical alloying** (e.g., high-energy ball milling) of the intrinsically brittle active materials with **ductile metallic additives** (e.g., Cu, Al, Zn, or Ni). This approach aims to modify the powder morphology and increase the overall ductility of the feedstock, thereby enhancing deposition efficiency.

As a secondary approach, we will explore the incorporation of **polymeric binders** into the feedstock blend. This would serve to bridge the active material particles and improve structural integrity, facilitating the desired mass loading.

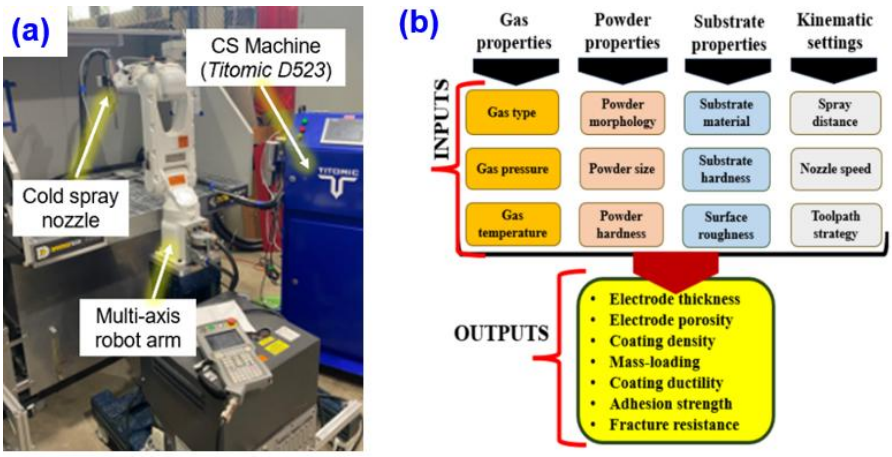

***Figure 4. (a)** Robotic CSAM cell in PI Akin's lab at RPI: **(b)** CSAM process parameters to be studied.*

Task 3: Electrochemical characterization for cell-level validation *(University-led)*

The electrochemical performance of CSAM-based anodes and cathodes will be systematically evaluated in a half-cell format against Li-metal using an *Arbin BT2000* tester (see **Figure 5**). Cells will be assembled with a *Celgard 2340* separator and *~1M $LiPF_6$ in 1:1 mixture of ethylene carbonate and diethyl carbonate* electrolyte. Cyclic voltammetry (CV) will identify redox peaks, while galvanostatic charge-discharge cycling will quantify specific capacity (volumetric and gravimetric), coulombic efficiency, and capacity retention. Following low-rate baseline testing, we will assess high-rate capability to evaluate fast-charging potential. To investigate the structural impact of lithiation, cycled electrodes will be harvested for ex-situ analysis; SEM and XRD will assess morphological stability and intercalation mechanisms, while EDS and XPS (with depth-profiling) will map Li-distribution and characterize the thickness and composition of the solid-electrolyte interface (SEI). Finally, reaction kinetics will be resolved via the Galvanostatic Intermittent Titration Technique (GITT) and Electrochemical Impedance Spectroscopy (EIS) to determine ion diffusion coefficients and deconvolute charge transfer, electrolyte, and SEI resistances.

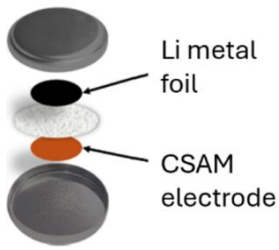

***Figure 5.** Coin-cell setup.*

(max 10 pages)

Task 4: Full-cell demonstration and testing *(Joint)*

To demonstrate commercial viability, it is essential to demonstrate full-cell devices with the CSAM anode and cathode electrodes. Successful full-cell testing will require proper capacity matching between the anode and cathode. To achieve this, we will adjust the area of the electrodes to achieve an N/P ratio (ratio of anode capacity to cathode capacity) of ~1.1. Maintaining an N/P ratio > 1, ensures that Li ions from the cathode can be fully accommodated by the anode, without the risk of Li plating or dendritic outgrowth.

Initial full-cell studies will be conducted at the coin-cell level. Subsequently, larger-format pouch cells will be fabricated at the Center for Future Energy Systems (CFES) at RPI. These pouch cells will serve as a Minimum Viable Prototype (MVP) for evaluating electrochemical performance and conducting preliminary techno-economic (cost) analyses. **Guidance from our Korean partner will be crucial for the proposed pouch cell MVP demonstration.**

Task 5: Techno-economic and scalability analysis *(Industry-led)*

The results from Tasks 1-4 will enable us to evaluate the manufacturing feasibility and viability of CSAM for LIB electrodes. However, its techno-economic viability for large-scale industrial deployment is crucial and **guidance from our Korean partner will be vital**. The **proposed techno-economic** and **scalability assessment** under this task will include:

- Process throughput and productivity analysis, including deposition rate, achievable electrode thickness per pass, and cycle time.
- Cost modeling of the CSAM-based dry electrode process, accounting for materials, energy use, labor, and equipment utilization.
- Scalability assessment, addressing multilayer deposition strategies, process repeatability, yield, and defect tolerance at increased electrode area.
- Benchmarking against traditional slurry-coating, and state-of-the-art wet and dry electrode manufacturing routes, highlighting advantages, limitations, and cost/performance trade-offs.
- Identify technical bottlenecks and recommend process modifications for industrial deployment.
- Develop cost models including capital expenditure and operational expenditure for CSAM.
- Perform sensitivity analysis to identify key cost drivers and their impact on total cost.
- Compare cost metrics (e.g., cost per Kg, cost per kWh) against conventional anodes and cathodes

Project timeline

Task	Topic	Year-1	Year-2	Year-3
1	Material feasibility & parameter optimization	----------		
2	Dry coating of industry relevant electrode materials	----------		
3	Electrochemical characterization for cell-level validation	----------		
4	Full cell demonstration and testing	----------		
5	Techno-economic and scalability analysis		----------	
	Interaction with Korean partner company	----------		

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute

- ➔ The principal investigator (**Prof. Semih Akin**) brings expertise in cold spray additive manufacturing (CSAM), advanced manufacturing, and surface engineering. He pioneered the application of CSAM in emerging areas such as flexible and printed electronics, and energy devices, where solid-state deposition enables the integration of functional materials on temperature-sensitive substrates [15-17]. He is also expert in the fundamental mechanistic modeling of CSAM process.
- ➔ The co-principal investigator (**Prof. Nikhil Koratkar**) brings expertise in battery materials and electro-chemistry. He has worked in the battery space for over 15 years and has published over 260 archival journal papers with > 41,000 citations (H-index = 96). He is a Fellow of the ASME, RSC, AAAS and APS.
- ➔ Two Ph.D. students at RPI will work on this project. One student trained in CSAM by Prof. Akin will perform the experimental work (Task 1-2), while the other student trained by Prof. Koratkar will carry out the battery experiments in collaboration with our Korean partner company (Tasks 3-4).

(max 10 pages)

	The techno-economic analysis (Task 5) will be conducted jointly by the entire team including our Korean partner company. ➔ Prof. Akin will be responsible for coordinating the research project. He will schedule weekly meetings for all participants to achieve synergy between material characterization, LIB manufacturing and electro-chemical testing. He will also provide regular feedback to our Korean partners to ensure that any course corrections (if needed) can be made in a timely manner. ➔ All the required manufacturing, material synthesis, characterization and manufacturing facilities are available to the investigators on the RPI campus. Specifically, RPI's **Center for Smart Convergent Manufacturing Systems (CSCMS)** provides all required manufacturing capabilities. The **RPI's Materials Research Center (MRC)** are well-equipped for mechanical and microstructural characterization. Together, these facilities provide comprehensive support for the proposed research and ensure seamless collaboration with Korean SME partners.
Expected R&R of Korean SME Partner	The Korean SME is expected to play a central role in driving the technical development, industrial implementation, and collaborative execution of the proposed project. Key responsibilities include: ➔ Provide domain expertise in LIB manufacturing process, materials, and production workflow optimization. ➔ Pouch cell testing to demonstrate a minimum viable prototype (MVP) full-cell device. ➔ Integration of CSAM-produced electrodes into existing LIB cell architectures. ➔ Perform durability and reliability assessment under operational conditions aligned with industrial requirements. ➔ Manufacturing scalability and cost analysis to establish commercial viability of the proposed technology. ➔ Analyze manufacturing metrics (i.e., throughput, cost effectiveness). ➔ Allocate skilled personnel to support development, testing, and data analysis.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	This project is expected to lead to significant technological advancements, as outlined below: ➔ Translate CSAM from a laboratory-scale demonstration to a robust, industrially deployable process for producing solvent- and binder-free LIB electrodes. ➔ Elimination of N-Methyl-2-Pyrrolidone (NMP) in battery processing. NMP is notorious for its health risks (reproductive toxicity) and high energy cost for drying. ➔ A clear pathway for integrating CSAM-enabled electrodes into rechargeable LIBs. ➔ Accelerating the development of advanced battery manufacturing capabilities across the US and Korea to reduce reliance on solvent-and binder-based electrode manufacturing. **Potential impacts:** ➔ High-density, defect-minimized electrodes with enhanced performance, durability, and reliability. ➔ Elimination of solvent-handling steps and reduction of overall production time through a **single-step electrode manufacturing process**. ➔ Removal of toxic solvents (e.g., NMP) and energy intensive drying processes, supporting more **sustainable, low-emission manufacturing**. ➔ Strengthened advanced battery manufacturing ecosystems in both countries.

6. References

(1) Wood, David L., et al. "Technical and economic analysis of solvent-based lithium-ion electrode drying with water and NMP." *Drying technology* 36.2 (2018): 234-244.

(2) Jain, R., Lakhnot, A. S., Bhimani, K., Sharma, S., Mahajani, V., Panchal, R. A., ... & Koratkar, N. (2022). Nanostructuring versus microstructuring in battery electrodes. *Nature Reviews Materials*, *7*(9), 736-746.

(3) Li, W., Yang, K., Yin, S., Yang, X., Xu, Y., & Lupoi, R. (2018). Solid-state additive manufacturing and repairing by cold spraying: A review. *Journal of materials science & technology*, *34*(3), 440-457.

(4) Wang, X., Feng, F., Klecka, M. A., Mordasky, M. D., Garofano, J. K., El-Wardany, T., ... & Champagne, V. K. (2015). Characterization and modeling of the bonding process in cold spray additive manufacturing. Additive Manufacturing, 8, 149-162.

(5) Champagne, V., & Helfritch, D. (2016). The unique abilities of cold spray deposition. *International Materials Reviews*, *61*(7), 437-455.

(6) Vaz, Rodolpho Fernando, Andrea Garfias, Vicente Albaladejo, Javier Sanchez, and Irene Garcia Cano. "A review of advances in cold spray additive manufacturing." *Coatings* 13, no. 2 (2023): 267.

(max 10 pages)

(7) Zhang, Zhe, Dongmei Han, Min Xiao, Shuanjin Wang, Yuhong Feng, Sheng Huang, and Yuezhong Meng. "New potential substitute of PVDF binder: Poly (propylene carbonate) for solvent-free manufacturing high-loading cathodes of LiFePO4| Li batteries." *Ionics* 29, no. 10 (2023): 3895-3906.

(8) Ryu, Minje, Young-Kuk Hong, Sang-Young Lee, and Jong Hyeok Park. "Ultrahigh loading dry-process for solvent-free lithium-ion battery electrode fabrication." *Nature communications* 14, no. 1 (2023): 1316.

(9) Ludwig, B., Zheng, Z., Shou, W., Wang, Y., & Pan, H. (2016). Solvent-free manufacturing of electrodes for lithium-ion batteries. *Scientific reports*, *6*(1), 23150.

(10) Kirsch, D. J., Lacey, S. D., Kuang, Y., Pastel, G., Xie, H., Connell, J. W., ... & Hu, L. (2019). Scalable dry processing of binder-free lithium-ion battery electrodes enabled by holey graphene. *ACS Applied Energy Materials*, *2*(5), 2990-2997.

(11) Tao, R., Steinhoff, B., Sun, X. G., Sardo, K., Skelly, B., Meyer III, H. M., ... & Li, J. (2023). High-throughput and high-performance lithium-ion batteries via dry processing. *Chemical Engineering Journal*, *471*, 144300.

(12) Kwon, K., Kim, J., Han, S., Lee, J., Lee, H., Kwon, J., ... & Choi, J. (2024). Low-Resistance LiFePO4 Thick Film Electrode Processed with Dry Electrode Technology for High-Energy-Density Lithium-Ion Batteries. *Small Science*, *4*(5), 2300302.

(13) Park, G., Kim, H. S., & Lee, K. J. (2023). Solvent-free processed cathode slurry with carbon nanotube conductors for Li-ion batteries. *Nanomaterials*, *13*(2), 324.

(14) Liang, Z., Li, T., Chi, H., Ziegelbauer, J., Sun, K., Wang, M., ... & Ban, C. (2024). Solvent-free manufacturing of lithium-ion battery electrodes via cold plasma. *Energy & Environmental Materials*, *7*(1), e12503.

(15) Akin, Semih, Seungjun Lee, Seunghwan Jo, Duygu Gazioglu Ruzgar, Karthick Subramaniam, Jung-Ting Tsai, and Martin Byung-Guk Jun. "Cold spray-based rapid and scalable production of printed flexible electronics." Additive Manufacturing 60 (2022): 103244.

(16) Akin, Semih, Taehoo Chang, Sk Shamim Hasan Abir, Young Won Kim, Shujia Xu, Jongcheon Lim, Yuseop Sim et al. "One-step fabrication of functionalized electrodes on 3D-printed polymers for triboelectric nanogenerators." *Nano Energy* 129 (2024): 110082.

(17) Kim, Young Won, Semih Akin, Huitaek Yun, Shujia Xu, Wenzhuo Wu, and Martin Byung-Guk Jun. "Enhanced performance of triboelectric nanogenerators and sensors via cold spray particle deposition." *ACS Applied Materials & Interfaces* 14, no. 41 (2022): 46410-46420.

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization